

Antarctica Bharati Research Station



Doppler Weather Radar

High-Performance Computing Facility of MoES to enhance weather and forecasting services have yielded nearly fifty-fold economic benefits to farmers, livestock rearers and fisherfolk, especially ~10.7 million below-poverty-line agricultural households and 0.53 million BPL fisherfolk households over five years, as per an assessment by the National Council of Applied Economic Research, New Delhi.

MoES runs successful Antarctic, Arctic, Southern Ocean, and Cryosphere programs. India has permanent research bases in Earth's poles: Himadri in the Arctic; Dakshin Gangotri (1983), Maitri (1988), and Bharti (2012) in the Antarctic, and Himansh in the Himalayas. Maitri and Bharati are operational. The ministry has concluded 42 scientific expeditions to the Antarctic, including during the pandemic. With ISRO (Indian Space Research Organisation), the Antarctica Ground Station for Earth Observation Satellites (AGEOS) at Bharati station in Antarctica facilitates data acquisition from Indian remote sensing satellites. India has an Arctic policy and the Antarctic Act. It hosted the Scientific Committee on Antarctic Research for the first time in 2022, establishing a vital scientific and knowledge foothold in international ties and cooperation in polar sciences.

MoES maintains a vast network of 158 seismological observatories as part of the National Seismological Network, enabling the country to monitor earthquakes in real-time 24/7. Earthquake vulnerability assessment or microzonation of cities is done to aid better urban planning. The country's only scientific deep drilling in Koyna, Maharashtra, is conducted by the BGRL to study reservoir-triggered earthquakes.

The ministry also provided an integrated flood warning system for Mumbai and Chennai; technologies for beach restoration and generating potable water from seawater; skill training for open sea age culture; and facilitates oil spill alerts; flash flood guidance services for India, Bangladesh, Bhutan, Nepal and Sri Lanka; research vessels and cruises through Sagar Nidhi, Sagar Manjusha, Sagar Tara and Sagar Anveshika; winter fog forecasts for IGI airport, New Delhi; and marine and coastal water monitoring.



Matsya Samudrayaan

MoES successfully launched India's flagship manned deep ocean mission 'Samudrayaan' under the Deep Ocean Mission, as part of which a self-propelled manned submersible called 'Matsya' will be developed.

MoES successfully launched India's flagship manned deep ocean mission 'Samudrayaan' under the Deep Ocean Mission, as part of which a self-propelled manned submersible called 'Matsya' will be developed. The Matsya would carry three crew members to a water depth of 6000 meters in the ocean with a suite of scientific sensors and tools for deep ocean exploration with an endurance of 12 hours of operational period and 96 hours in case of emergency. Underwater locomotion trials 5270 meters deep have shown promising progress. Deep Ocean Mission consists of six major components: development of technologies for deep sea mining and manned submersible called Matsya and tools and an integrated system for underwater mining 'Varaha' for extracting precious polymetallic nodules from the central Indian Ocean; ocean climate change advisory services and awareness about Earth system science; innovations for exploring and conserving deep-sea biodiversity; deep ocean survey to study and identify potential sites of multi-metal hydrothermal sulphides mineralisation along the Indian Ocean mid-oceanic ridges; harness energy and freshwater from the ocean; and establish an advanced Marine Station for Ocean Biology. This mission is set to boost India's blue economy.

To ensure that MoES services reach citizens directly, the ministry has user-friendly mobile applications, most of them also in vernacular languages, available free of cost for both Android and iOS platforms. The mobile apps include Damini (lightning), Meghdoot (weather and climate-related agro-met advisories for farmers and livestock-rearers), Samudra (ocean-related data and services including fishing zone advisories, alerts for extreme ocean events, ocean state, etc., for fisherfolk and coastal communities), Bhookamp (earthquakes), Safar (air quality), Sarat (Search and Rescue Aid Tool for objects or persons lost at sea), SVAS (Small Vessel Advisory and Forecast Services System), CBAS (Coral Bleaching Alert System), ABIS (Algal Bloom Information Service), SSFS (Swell Surge Forecast System), D-COIN (Digital Coast-India for data on coastal and marine pollution, litter, shoreline changes, coastal hazards, etc.), Thoondil (ocean state information in regional languages) and Mausam (weather). Information is also disseminated through print media, academic publications, and digital platforms such as websites, web portals, and social media.

The outreach arm of MoES is strong. The ministry facilitates the annual International Earth Science Olympiad to enhance students' interest and learning of Earth sciences at the school level and generate public awareness. The ministry proactively pursues awareness, capacity-building and international cooperation through programs such as student interaction, workshops, seminars, conferences, funding, meetings, etc., and makes documentaries and videos available. The first-of-its-kind, nationwide and citizen-led campaign called Swachh Sagar Surakshit Sagar in 2022 and 2023 helped clean more than 150 Indian beaches, envisioned to improve ocean health through mass awareness and collective action. As part of India's G20 Presidency, MoES spearheaded the 'Oceanic Track' under the Environment, Climate and Sustainability Group and organised a Research and Innovation Gathering themed Scientific Challenges and Opportunities for a Sustainable Blue Economy.

MoES has ambitious plans for activities under all schemes. Annual Indian scientific expeditions to the Arctic have commenced in 2023. Procurement of a High-Performance Supercomputing Facility of 18 petaflops, three times faster than the existing one, is finalised. The weather, climate, and seismological observation systems network is set to widen. Deep Ocean Mission is on track. MoES remains steadfast and committed to fulfilling its mandate of science for public service and nation-building.

BHOOKAMP DAMINI MAUSAM MEGHDOOT SAFAR SAMUDRA THOONDIL